

Injae Jun

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Education

BS	Seoul National University (SNU) , Mechanical Engineering	Mar 2020 – Present (Expected Aug. 2026)
	• GPA: 3.77/4.0 • Coursework: Mechatronics (A+), Mechanical Product Design (A+) • 2 years of absence due to military service	

Research Interests

- Robotics (Soft Robotics, Bio-inspired Mechanisms)
- Exoskeletons & Prosthetics (Novel Actuation Mechanisms, Human-Centered Design, Control Strategies)

Research Experience

SOFT ROBOTICS & BIONICS LAB (SRBL)	Jul 2024 – Present Research Link 
Research Intern – Advisor: Prof. Yong-Lae Park	
• Developed a compact thumb opposition wearable for patients with thenar muscle dysfunction, integrating tendon-driven and rack-pinion actuators with ergonomic hardware design. • Fabricated and integrated EGaIn-based stretchable sensors for motion sensing.	
Healthcare Robotics Lab (HeRo Lab)	Jan 2025 – Present
Research Intern – Advisor: Prof. Amy Kyungwon Han	
• Researched twisted string actuators and designed a novel bistable transmission mechanism to enhance their force and speed output. • Investigated human hand musculature and grasping biomechanics, applying findings to the design and prototyping of a prosthetic hand system.	

Projects

Auto Balancing Case	Project Link 
• Prototyped a self-balancing luggage system that reduces wrist load using load-cell sensing, differential actuation, and reinforcement learning control. • Validated robustness through PhysX-based simulation and Sim2Real transfer to hardware. • Tools Used: Fusion 360, Arduino, IsaacSim, RSL-RL (PPO), Python	
Auxetic Structure Optimization for Crash Box	Project Link 
• Modeled auxetic unit cells and optimized their geometry using Grid Search, CMA-ES, and FEA to maximize crash energy absorption. • Conducted compression and impact tests on 3D-printed TPU prototypes, achieving up to 26.5% improvement in energy absorption. • Tools Used: MATLAB, Python, Finite Element Analysis, CMA-ES, 3D Printing	
Reverse Engineering and Improvement of the FCX24 Lemur	Project Link 
• Analyzed chassis strength with ANSYS, drivetrain/transmission gear ratios, and suspension dynamics of an RC car model. • Improved design by shifting COM and tuning suspension damping; fabricated a	

custom damper and experimentally measured damping coefficient.

- Tools Used: ANSYS, Fusion 360, MATLAB, Zaber, Force Gauge

Multivariate Time Series Forecasting using XGBoost & Neural Networks

[Project Link](#)

- Predicted next-day oil temperature on the ETT dataset by redefining the target as daily difference, achieving **top 1%** accuracy among 200 participants.
- Tools Used: Python, XGBoost, PyTorch, Scikit-learn, Pandas, NumPy

Leadership / Extracurricular activities

SNU Computer Study Club (SCSC)

Mar 2020 – Aug 2020

- Participated in Python and C programming study groups; developed a Minesweeper game project in C.

Run to You – SNU Baja/Formula 1 Team

Apr 2020 – Oct 2021

Team Leader – Baja Chassis Team

- Led the team in designing and manufacturing an optimized chassis for vehicle performance, and participated in a national student automobile competition(KSAE).

Republic of Korea Army

Mar 2022 – Sep 2023

Squad Leader – Sergeant, 21st Aviation Brigade

- Led a 7-soldier squad while supporting maintenance and operation of utility helicopters, ensuring unit readiness and participation in multiple field training exercises.

DB Business Management Competition

Jan 2024

1st Prize – Business Performance

- Gained experience in corporate management (inventory, HR, finance) and achieved top performance in a business simulation competition.

Honors and Awards

Busan Metropolitan City Mayor's Award (3rd Place)

November 2025

COSS MCU Application Competition

2025 CO-SHOW

Next-Generation Engineer Award

September 2025

Institute for Promotion of Engineering and Science of Korea

IPESK

Merit-based Scholarship

Spring 2025

Department of Mechanical Engineering

Seoul National University

Skills

Programming: Python, Matlab, C/C++, PyTorch

Software & Hardware: CAD (SolidWorks, Blender), ANSYS, Arduino, Raspberry Pi, 3D Printing, Laser Cutting

Languages: Korean (Native), English (Fluent, TOEFL 107)